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Friction

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FRICTION

- When two objects are in contact, there is often a tendency for them to slide relative to one another.
- When this is the case, **friction forces** that resist the sliding motion develop on the contact surfaces between the objects.
- A number of factors influence how large the friction forces can be, and the field of study that addresses this topic is called tribology , which is derived from the Greek Word *tribos* , which means *rubbing*.
- Early experiments in the study of friction forces were done by Leonardo da Vinci (1452–1519) and Guillaume Amontons (1663–1705).

FRICTION

- **Leonardo da Vinci** found that friction force and normal force were proportional.
- **Guillaume Amontons** lubricated various combinations of iron, copper, lead, and wood with pork fat. He found the friction force F during sliding and the normal force N were related by $F \approx N/3$.
- He also found that the friction force was independent of the apparent area of contact.
- Charles Augustin de **Coulomb** in 1781 confirmed the findings of Amontons and distinguished between **static** and **kinetic** friction.

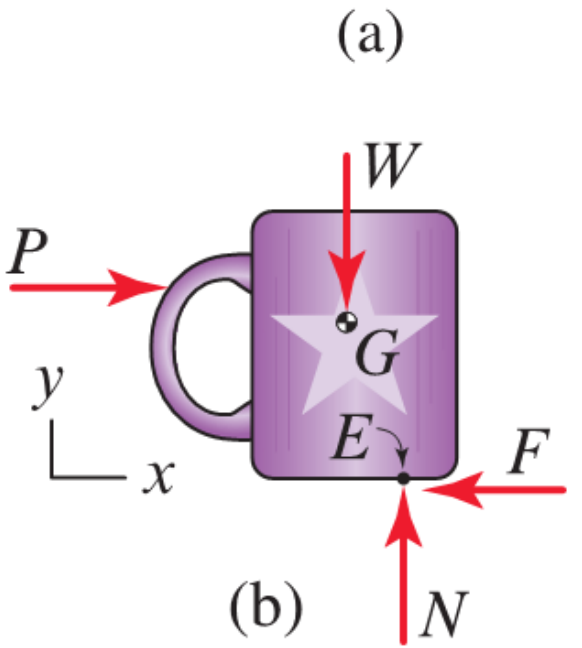
FRICTION - A Simple Experiment

- An empty coffee cup with **weight W** rests on a table (shown in top-right).



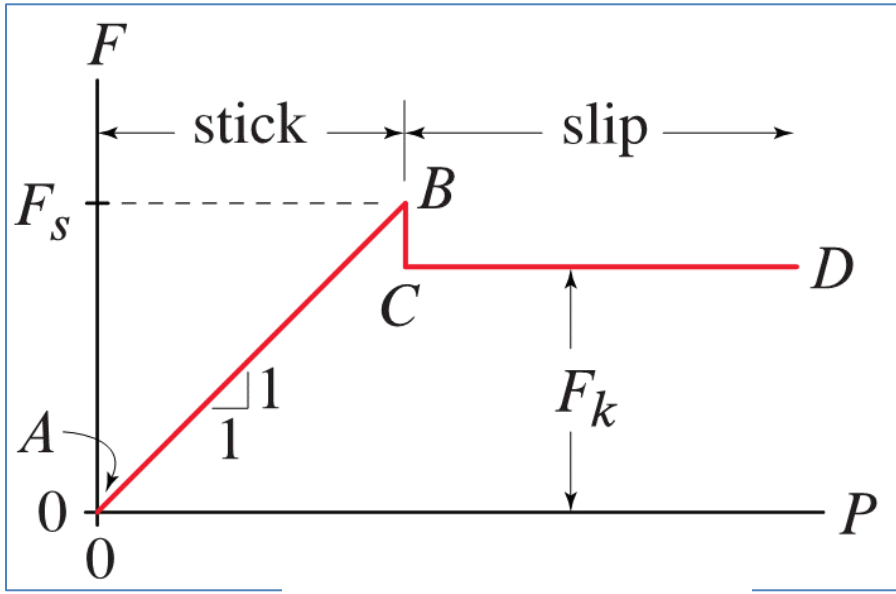
- Imagine the fingers were used to apply a slowly increasing horizontal **force P** to the cup, starting **from a value of zero**, so as to **slide the cup to the right**.

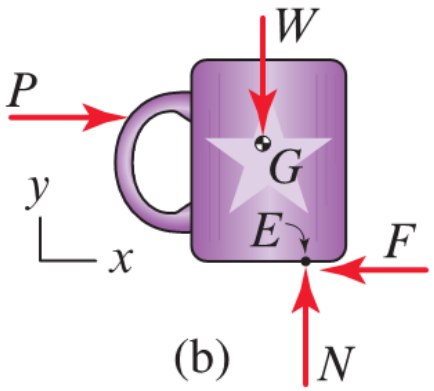
- Between the cup and table, in addition to the **normal force N**, a **friction force F** develops, as shown in the FBD.



FRICTION - A Simple Experiment

- The relationship between **P** and the **friction force F** that one is likely to observe is shown.
- In region **AB** of this figure, the cup undergoes no motion (this is often called **stick**) and **P = F**.
- At point **B** , motion is **impending**, i.e., that is, slip is about to occur.
- For values of **P beyond point B**, the cup-table interface is not capable of supporting friction forces that are high enough to provide equilibrium.
- In this regime, **P>F** and sliding occurs.



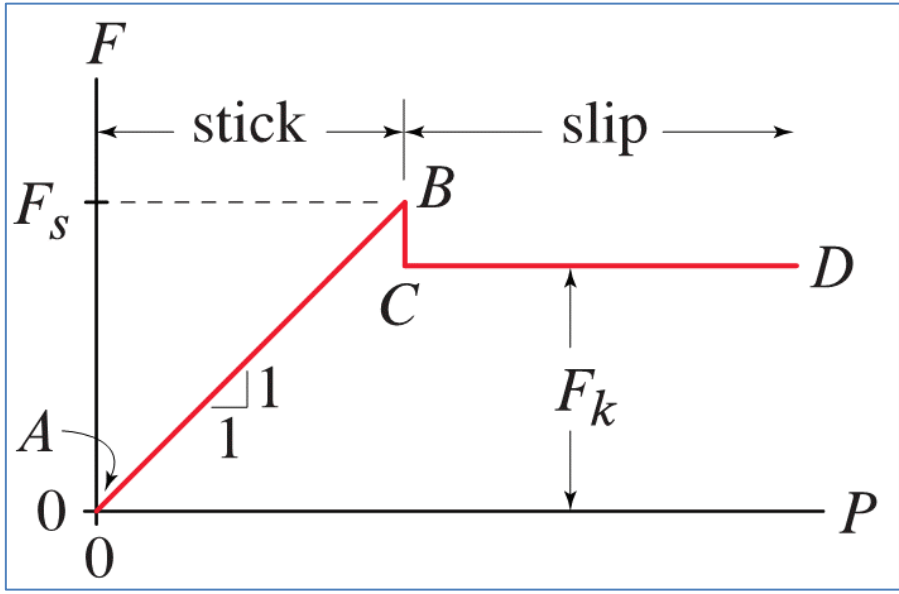


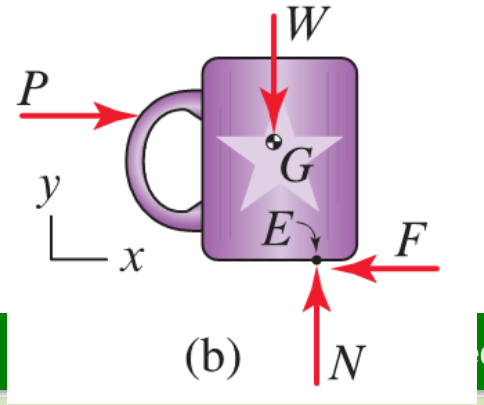
FRICTION - A Simple Experiment

- If the experiment is repeated using a cup filled with enough coffee to **double the weight** of the original empty cup, **the force at which sliding starts is found to be approximately doubled.**

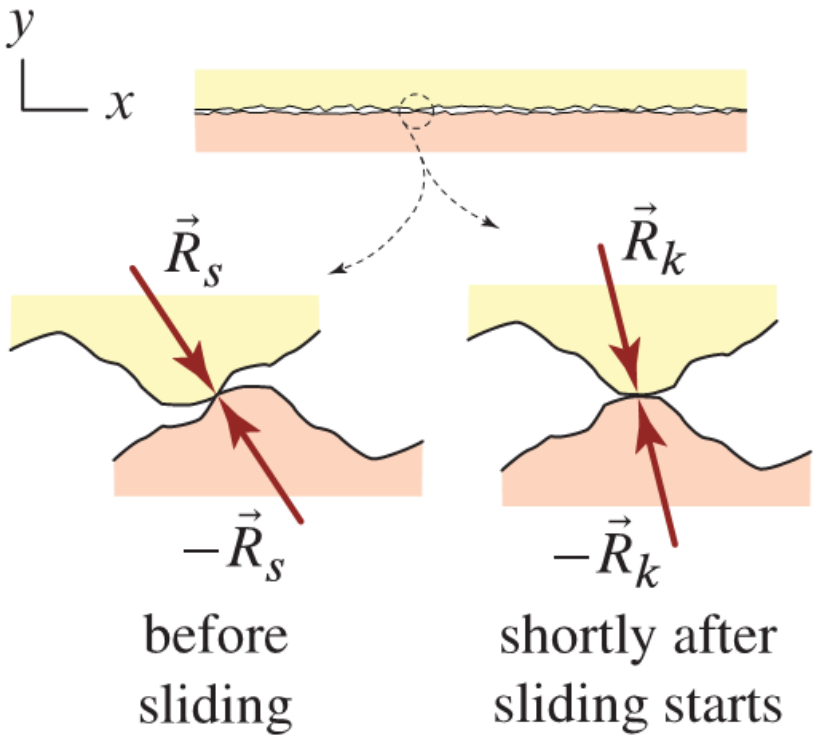
- Typically, the friction force $\mathbf{F_s}$ at which sliding starts is higher than the friction force $\mathbf{F_k}$ for sustained sliding.

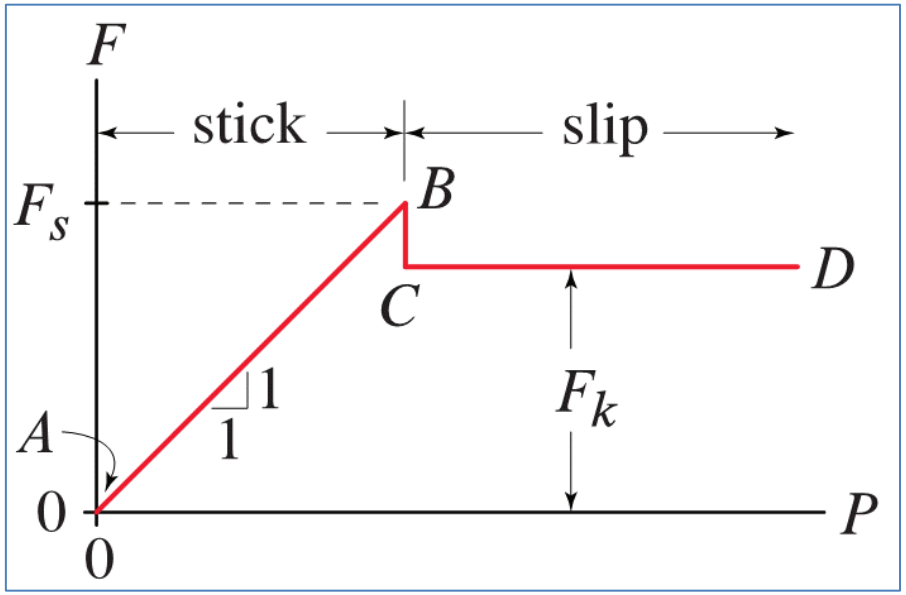
- these values are respectively called the **static friction force** and the **kinetic friction force**.





Why is $F_s > F_k$?

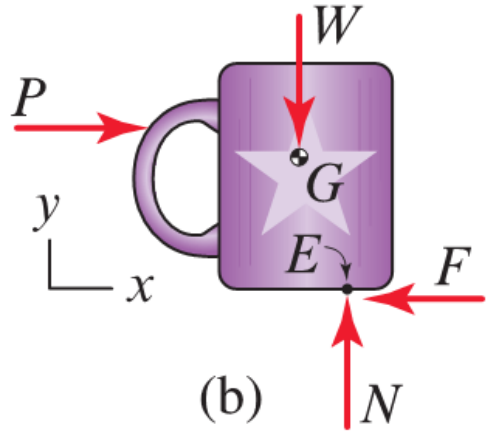




Explained in terms of asperities.

Coulomb's law of friction

- Coulomb's law presents a model that describes the relationship between the **normal force** and **friction force** between two contacting surfaces.



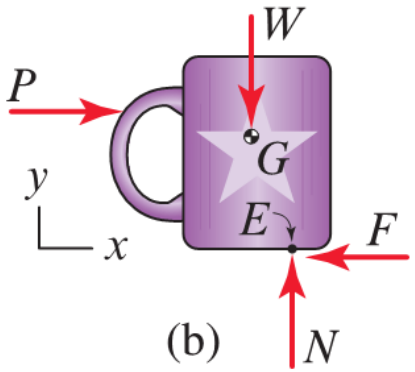
$$|F| \leq \mu_s N$$

- ❑ (<) **before sliding and**
- ❑ (=) **for impending motion.**
- ❑ $|F| = \mu_k N$ **after sliding begins.**

- **N** is the **normal force** between the two surfaces;
- **F** is the **friction force** between the two surfaces. Its direction always opposes relative motion between the surfaces;
- μ_s is called the **coefficient of static friction**;
- μ_k is called the **coefficient of kinetic friction.**

Coulomb's law of friction

- $|F| \leq \mu_s N$: (<) **before sliding and** (=) **for impending motion.**
- $|F| = \mu_k N$: **after sliding begins.**



- Coulomb's Law is written in its compact form: $|F| \leq \mu N$
- Where it is understood that the coefficient of static friction is used prior to sliding and when motion is impending, and the coefficient of kinetic friction is used once motion starts along with the = sign.
- Generally, due to the lack of reliable values of coefficients of friction, it is often assumed that $\mu_s = \mu_k$.

Coefficients of friction

- The friction force between two contacting surfaces is controlled by an enormous number of factors. These include:
  - ✓ Compressive force
  - ✓ The combination of contacting materials
  - ✓ Surface roughness
  - ✓ Lubricants
  - ✓ The environments the surfaces are in
  - ✓ Temperature
  - ✓ Humidity
  - ✓ Etc.

Coefficients of friction for various combinations of contacting materials.

Material	Coefficient of friction[a]	
	On self	On mild steel
aluminum	0.8–1.2	0.5–0.6
iron	0.8–1.5	0.8–1.5
leaded bronze	–	0.2–0.4
gray cast iron	0.8–1.0	0.3–0.5
mild steel	0.7–0.9	–
rock (μ_s)	0.4–0.8	–
diamond	0.1–0.2	–
hard surface with liquid lubricant	0.02–0.05	–
hard surface with solid lubricant	0.05–0.15	–
tire rubber on pavement (μ_s)	0.4–1.4	
soft rubber on glass (μ_s)	4–10	

[a] Data taken from B. Bushan, *Introduction to Tribology,* John Wiley & Sons, New York, 2002; K. C. Ludema, *Friction, Wear and Lubrication,* CRC Press, Boca Raton, 1996; R. E. Goodman, *Introduction to Rock Mechanics*, John Wiley & Sons, New York, 1980.

- Unless otherwise stated, values reported are coefficients of kinetic friction for clean and unlubricated surfaces in a normal air environment at 20°C

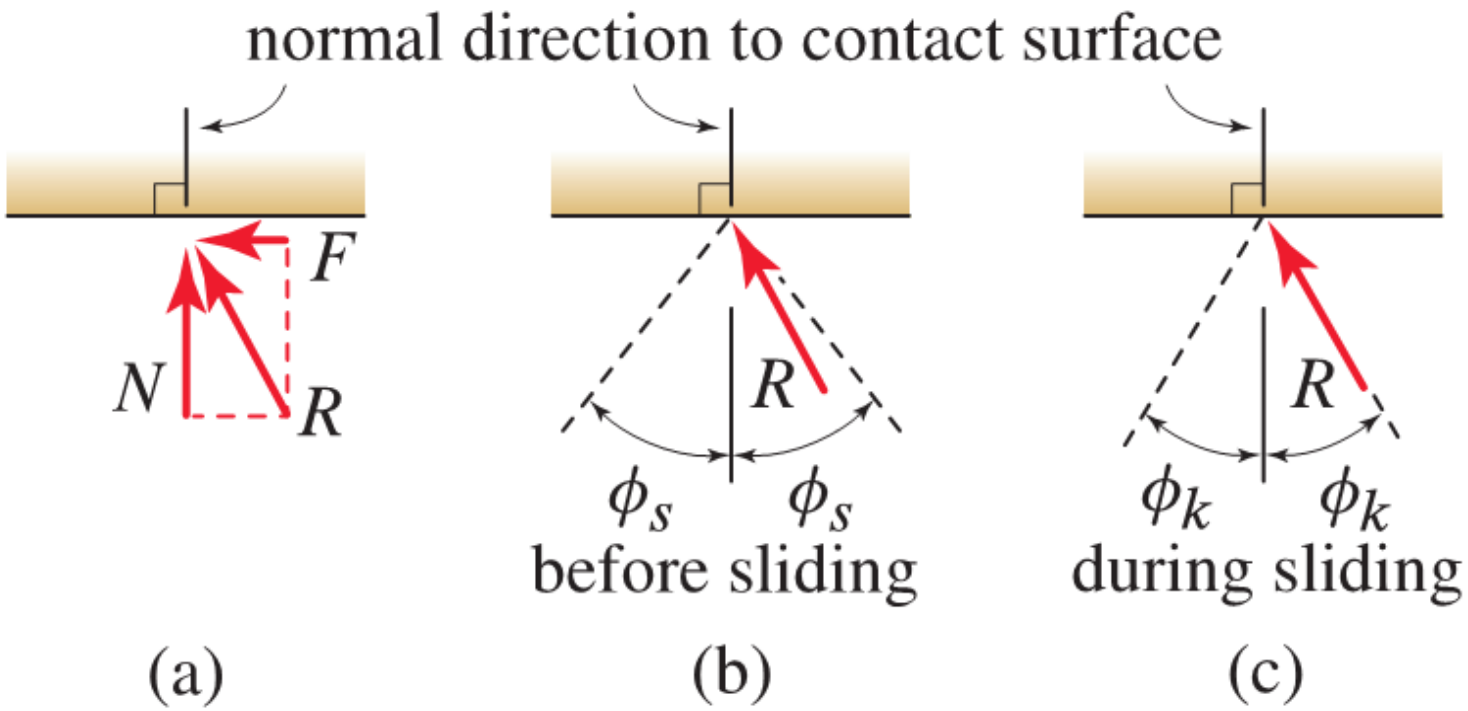
Dry Contact versus Liquid Lubrication

- Coulomb's law is best suited for dry contact.
- When surfaces have a liquid lubricant, hydrodynamic effects occur and Coulomb's law cannot be used.
- Where small amount of liquid lubricant is present between two contacting surfaces, it is still common to use Coulomb's law,

Angle of friction

- In some engineering disciplines, (e.g. geotechnical engineering), it is more common to characterize friction between surfaces by using the **angle of friction** rather than the coefficient of friction.
- The angle of friction is defined to be: $\emptyset = \tan^{-1} \mu$
  - Where μ is either the **static or kinetic coefficient of friction**.
- If a particular combination of contacting materials is known to have μ_s=0.35 ,then its angle of static friction is φ_s= 19.3º.
- The angle of friction has a straightforward physical interpretation.

Angle of friction



(c) During sliding, the resultant force lies on the cone defined by the angle of kinetic friction.

(a) **R** is the resultant of the normal force **N** and friction force **F** acting on a surface.

(b) The angle of static friction measured from the normal direction of the surface defines a cone. **No sliding** occurs if the resultant force lies within the cone. **Impending slip** occurs when the resultant force lies on the cone.